

3.1.6 Materials and their working properties

Students should know and understand the categorisation of the types and properties of the following materials.

3.1.6.1 Material categories

Papers and boards

Content	Potential links to maths and science
Students should have an overview of the main categories and types of papers and boards: papers including: • bleed proof • cartridge paper • grid • layout paper • tracing paper boards including: • corrugated card • duplex board • foil lined board • foam core board • ink jet card • solid white board.	Classification of the types and properties of a range of materials. Physical properties of materials related to use and knowledge applied when designing and making.

Natural and manufactured timbers

Content	Potential links to maths and science
Students should have an overview of the main categories and types of natural and manufactured timbers: hardwoods including: • ash • beech • mahogany • oak • balsa softwoods including: • larch • pine • spruce manufactured boards including: • medium density fibreboard (MDF) • plywood • chipboard.	Classification of the types and properties of a range of materials. Physical properties of materials related to use and knowledge applied when designing and making.

Metals and alloys

Content	Potential links to maths and science
Students should have an overview of the main categories and types of metals and alloys: ferrous metals including: • low carbon steel • cast Iron • high carbon/tool steel non ferrous metals including: • aluminum • copper • tin • zinc alloys including: • brass • stainless steel • high speed steel.	Classification of the types and properties of a range of materials. Physical properties of materials related to use and knowledge applied when designing and making.

Polymers

Content	Potential links to maths and science
Students should have an overview of the main categories and types of polymers: thermoforming including: • acrylic (PMMA) • high impact polystyrene (HIPS) • high density polythene (HDPE) • polypropylene (PP) • polyvinyl chloride (PVC) • polyethylene terephthalate (PET) thermosetting including: • epoxy resin (ER) • melamine-formaldehyde (MF) • phenol formaldehyde (PF) • polyester resin (PR) • urea-formaldehyde (UF).	Classification of the types and properties of a range of materials. Physical properties of materials related to use and knowledge applied when designing and making.

Textiles

Content	Potential links to maths and science
Students should have an overview of the main categories and types of textiles: natural fibres including: • cotton • wool • silk synthetic fibres including: • polyester • polyamide (nylon) • elastane (lycra) blended and mixed fibres including: • cotton/polyester woven including: • plain weave non-woven including: • bonded fabrics • felted fabrics knitted textiles including: • knitted fabrics.	Classification of the types and properties of a range of materials. Physical properties of materials related to use and knowledge applied when designing and making.

3.1.6.2 Material properties

Students should have an understanding of the working and physical properties of the materials in [Material categories](#) (page 16).

Material properties

Content	Potential links to maths and science
In relation to the main categories outlined above (not the specific materials identified), students should know and understand physical properties such as: • absorbency (resistance to moisture) • density • fusibility • electrical and thermal conductivity. In relation to the main categories outlined above (not the specific materials identified), students should know and understand working properties such as: • strength • hardness • toughness • malleability • ductility and elasticity.	Scientific vocabulary eg metals/non-metals and physical and chemical differences between them eg types and properties across a range of materials. Using materials eg composition of some important alloys eg selection of an alloy for enhanced durability in a particular design situation.